

# Eureka

**B.Tech Computer Science and Engineering, 8<sup>th</sup> Semester, 2021-22**



## Mentor

**Prof. Sudipta Dutta**

Assistant Professor

Dept. of CSE & IT

Bengal Institute of Technology, Kolkata

## Group Members

| Name                | MAKAUT Roll No. |
|---------------------|-----------------|
| Arunava De          | 12100118049     |
| Arkaprava Chowdhury | 12100118051     |
| Suman Maity         | 12100119122     |
| Sourav Kundu        | 12100119118     |
| Abhirup Kusari      | 12100119121     |



**BENGAL INSTITUTE OF TECHNOLOGY**

**1 GOVT. COLONY, TECH TOWN, DHAPA MANIPUR, ON BASANTI  
HIGHWAY, HADIA, KOLKATA-150**

## ***Acknowledgement***

*This project would not been a successful without the sincere cooperation and guidance of our mentor Prof. Sudipta Dutta, Assistant Professor, Dept of CSE & IT, Bengal Institute of Technology, Kolkata, who has provided us with useful resources and motivation through the various phases of this project which has prove to be crucial for the completion of this documentation.*

### **Signature of Group Members:**

-----  
Name: **Arunava De**  
MAKAUT Roll No.  
**12100118049**

-----  
Name: **Arkaprava Chowdhury**  
MAKAUT Roll No.  
**12100118051**

-----  
Name: **Suman Maity**  
MAKAUT Roll No.  
**12100119122**

-----  
Name: **Sourav Kundu**  
MAKAUT Roll No.  
**12100119118**

-----  
Name: **Abhirup Kusari**  
MAKAUT Roll No.  
**12100119121**

### **Signature of Mentor**

-----  
Prof. <Name of the mentor>  
Assistant Professor  
Dept. of CSE & IT  
Bengal Institute of Technology

### **Signature of HOD**

-----  
Dr. Shanta Phani  
Head of the Department  
Dept. of CSE & IT  
Bengal Institute of Technology

## TABLE OF CONTENTS

| Topic                                                                                                                          | Page Number |
|--------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1. <b>Scope of the Proposed System:</b> Goals, objectives and users                                                            |             |
| 2. <b>System Requirement Specification:</b>                                                                                    |             |
| 2.1. <b>Functional Requirements:</b>                                                                                           |             |
| 2.1.1. <b>Process Identification:</b> Process descriptions and Use Case                                                        |             |
| 2.1.2. <b>Logic Identification:</b> Per process inner logic specification<br>in Structured English or through Activity Diagram |             |
| 2.2. <b>Hardware Requirements</b>                                                                                              |             |
| 3. <b>System Design:</b>                                                                                                       |             |
| 3.1. <b>Class Diagram</b>                                                                                                      |             |
| 3.2. <b>Database Design</b>                                                                                                    |             |
| 4. <b>Future Work:</b> Plan for the Next Phase                                                                                 |             |
| 5. <b>References</b>                                                                                                           |             |

## 1. Scope of the Proposed System:

Eureka is a platform where you can get your future guidance, our vision is to provide the best education center for a student base of his/her provided criteria and help him/her to secure best possible future.

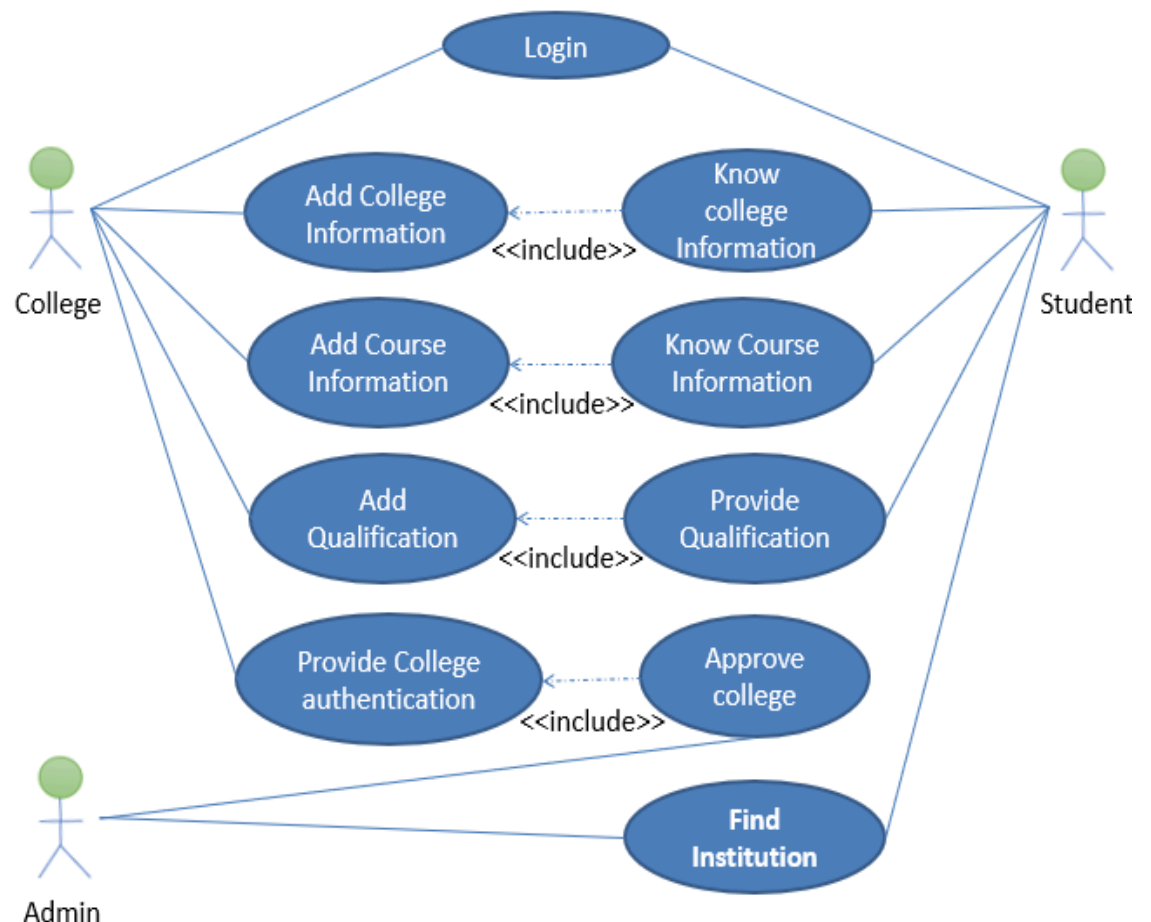
Our overall goal is to provide students a proper guidance regarding their future studies and admission in college. Our users will be mainly students and colleges/universities.

## 2. System Requirement Specification:

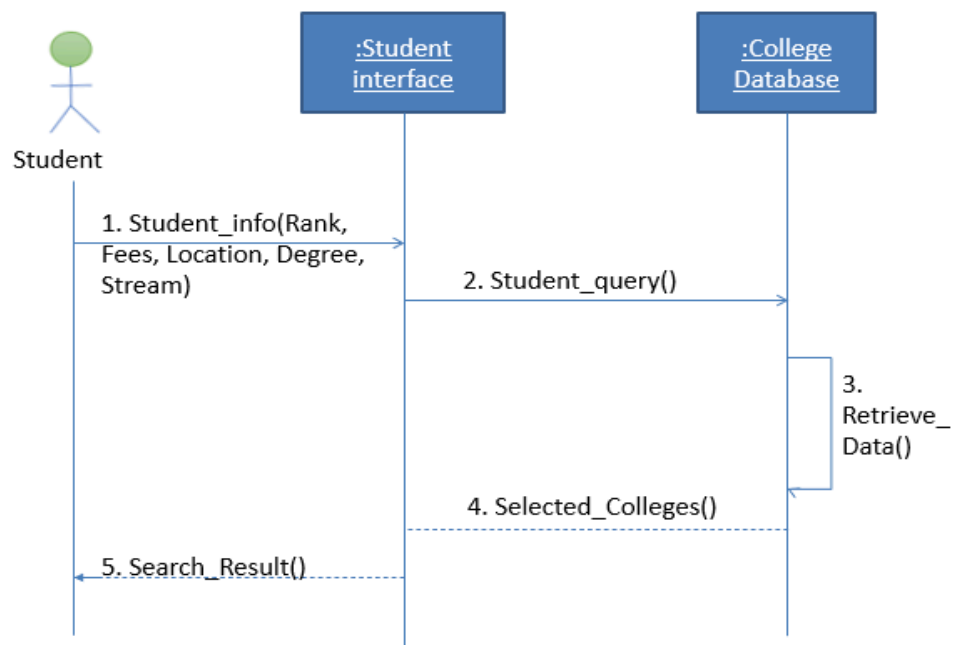
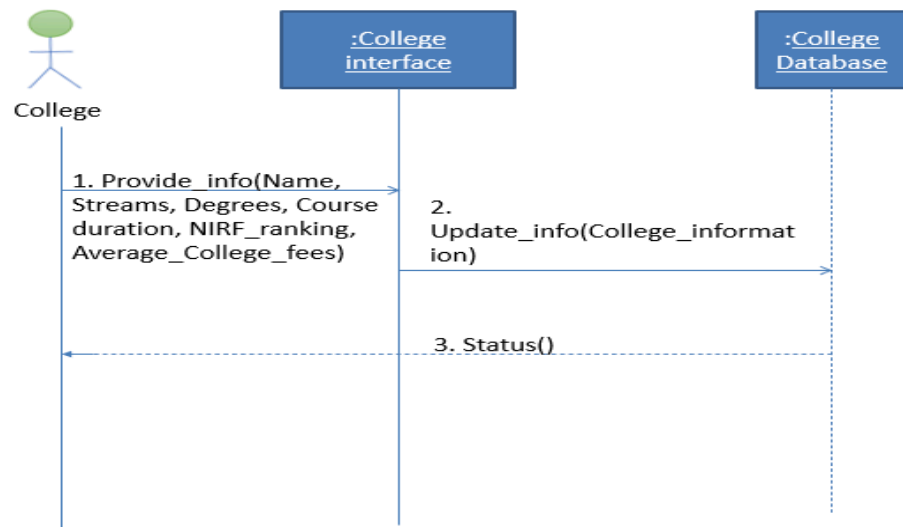
### 2.1 Functional Requirements:

#### 2.1.1

#### Add college information:



#### Find institution:



### 2.1.2 Logic Identification:

#### Characteristic of collages:

**College Details:** Name, Address

**College type:** government / Private / affiliated / autonomous

**Degree:** Agricultural, Agricultural Engineering, Nursing, Management, Medical, Para medical, Engineering (Civil, Mechanical, Electrical, Computer Science, Electronic, Information Technology), Music, Photography, Mass com, Pharmacy

**Stream:**

**Course duration:** duration of the total course

**Average fees:**

**Ratings:** 1-5

#### Characteristic of students:

##### Personal details:

Name, Guardians' Name, Address

##### Qualification:

Registration No, Name, Last examination, Rank/percentage (%)

##### Opting Degree:

(Bachelors) Agricultural, Nursing, Management, Medical, Para medical, Engineering, Music, Photography, Mass com

(Diploma) Pharmacy, Nursing, Engineering, Management

## **Characteristic of Admin:**

**Add register user (Student/College)**

**Review user information (Review the user information with company policy)**

**Change User Information:**

E-mail, Password, Student Name/College Name, etc.

**Find Institution (Find the best Institution of student's satisfaction)**

**Find students (Find the students of Institutions will want)**

### **2.2 Hardware Requirements**

**2.2.1** Processor: 2GHz or more

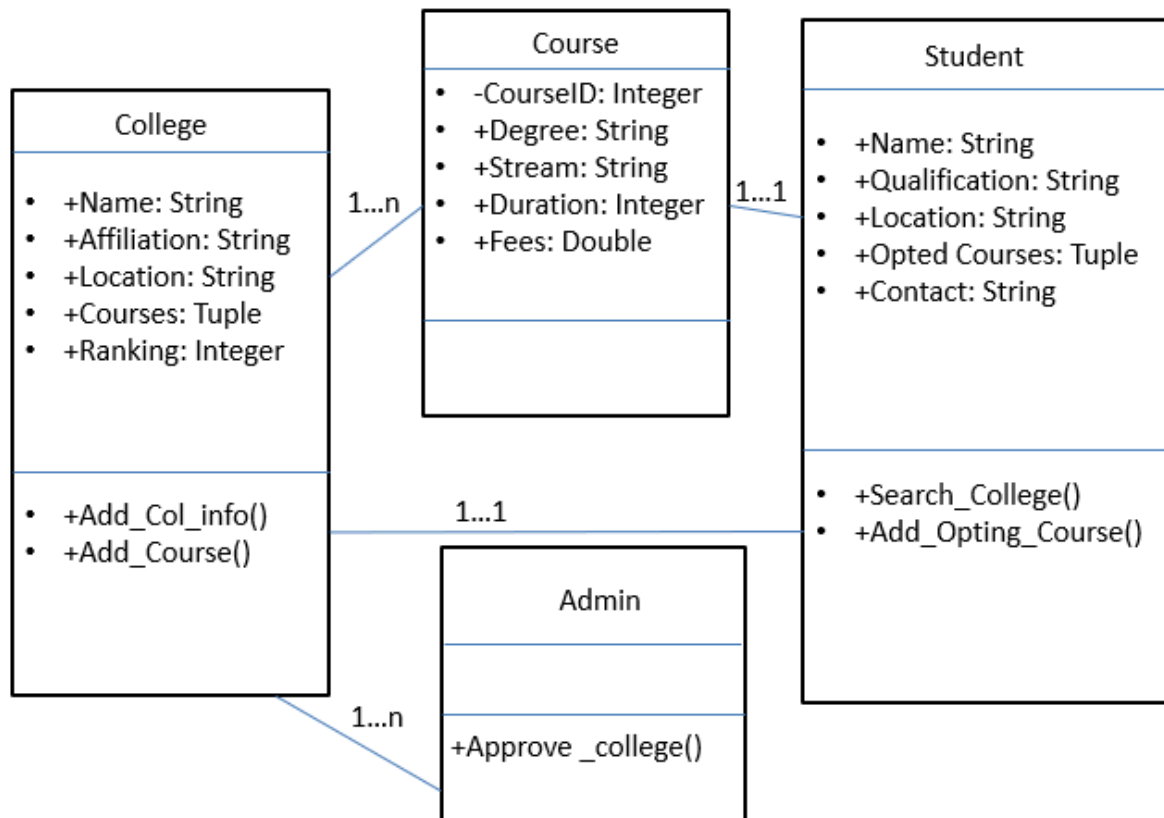
**2.2.2** Ethernet connection (LAN) OR a wireless adapter (Wi-Fi)

**2.2.3** Hard Drive: 64 GB or more

**2.2.4** Memory (RAM): 4 GB or above

## **3. System Design:**

### **3.1 Class Diagram**



### 3.2 Database Design

| Table Name: <b>STUDENT</b> |           |                                              |
|----------------------------|-----------|----------------------------------------------|
| Field Name                 | Data Type | Description                                  |
| NAME                       | String    | Name of the student                          |
| OPTED_COURSES              | Tuple     | Student preferring the courses (Multivalued) |



| Table Name: <b>UNIVERSITY</b> |           |                               |
|-------------------------------|-----------|-------------------------------|
| Field Name                    | Data Type | Description                   |
| <u>UNIV_ID</u>                | Integer   | Unique ID for each university |
| NAME                          | String    |                               |
| AFFILIATED_COLLEGES           | Tuple     | (Multivalued)                 |
| Degrees                       | Tuple     | (Multivalued)                 |
| Location                      | String    |                               |

| Table Name: <b>DEGREE</b> |           |                        |
|---------------------------|-----------|------------------------|
| Field Name                | Data Type | Description            |
| <u>DEGREE_NAME</u>        | String    |                        |
| STREAMS                   | Tuple     | (Multivalued)          |
| DURATION                  | Integer   | Duration of the Degree |

| Table Name: <b>COLLEGE</b> |           |                               |
|----------------------------|-----------|-------------------------------|
| Field Name                 | Data Type | Description                   |
| <u>COLL_ID</u>             | Integer   | Unique ID for each university |

|          |        |               |
|----------|--------|---------------|
| NAME     | String |               |
| Degrees  | Tuple  | (Multivalued) |
| Location | String |               |

| Table Name: COURSE |           |             |
|--------------------|-----------|-------------|
| Field Name         | Data Type | Description |
| <u>COURSE_ID</u>   |           |             |
| DEGREE             |           |             |
| DURATION           |           |             |
| STREAM             |           |             |

**4. Future Work:**

Our next goal is project design.

**5. References**

<https://collegedunia.com/>

<https://www.tutorialspoint.com/>